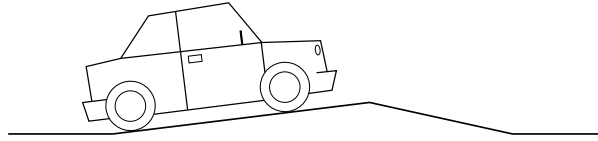


Exercise 4.1

Consider a car driving on a road, as shown below.



The mass of the car is 1000 kg and the drag force is linear in the velocity, with drag coefficient $b = 500 \text{ N/(m/s)}$. The throttle input to the system (considered to be a force) is delayed by 0.7 s before affecting the car. We intend to control the velocity of the car by use of the throttle input.



1. Tune a P-controller for the system using the Ziegler-Nichols tuning method, and subsequently ensure that the overshoot is below 2 % by modifying the controller gain on the basis of a step response of the system in MATLAB.



2. Tune a PI-controller for the system using the Ziegler-Nichols tuning method, and subsequently ensure a 5%-settling time below 3 s by modifying the controller gains on the basis of a step response of the system in MATLAB.



3. Assume that there is no input delay. Compute a symbolic expression for the closed-loop poles of the system with a PI-control, and place the closed-loop poles to obtain your desired system behavior (less than 2% overshoot and 5%-settling time below 3 s).
4. Simulate the closed-loop system on road conditions changing from flat to a 10° incline.